

Pinki Kumari

Department of Computer Science and Engineering

pinkikumari22.github.io

Education

Degree	Institution	CPI	Year
MTech	IIT Gandhinagar	8.13	2022 - 2024
BTech	VGI Microsoft, Greater Noida (AKTU)	8.33	2016 - 2020

Experience

- **Teaching Assistant, Indian Institute of Technology Gandhinagar**
 - Software Engineering & Testing (SET). [Jan - May 2024]
 - Computing (ES112): Designing and conducting lab tasks and checking assignments. [Aug - Dec 2023]
 - Aarohan (FP601): Managed orientation of 450+ students with a team of 25 TAs and was in the backend team, including five people. [July 2023]
 - Data-Centric Computing: Conducted labs and tutorials on topics of ML, EDA, Algorithms. [Jan - July 2023]
 - Computer Networks (CS433): Handled a class of 75+ students, conducted quizzes, checked assignments, and assessed final project presentations. [Aug - Nov 2022]
- **Summer Intern, ThirdAI Corp.** [June 1st - July 31st, 2023]
 - Utilized ThirdAI's Universal Deep Transformer (UDT) for pre-training, training, and fine-tuning LLMs on CPU instead of GPU.
 - Embedded evaluation system for benchmarking various models like BERT, Open AI, LLaMA, and BOLT.

Skill Summary

- **Language and Frameworks:** Python, SQL, HTML5, CSS, Kaggle, DrivenData, Git, Flask, VSCode, Hugging face
- **Data Science Libraries:** PyTorch, TensorFlow, Keras, Matplotlib, Seaborn, Scikit-Learn, Numpy, Pandas, JAX
- **General:** Machine Learning, Deep Learning, Natural Language Processing, Automatic Speech Recognition, Prompt Engineering

Projects

- **Automatic Speech Recognition(ASR) using Neural Tangent Kernels(NTKs)** [Jan 2023 - May 2024]
 - ASR on Amharic Audio Language that is build on neural tangent kernels (no need of neural networks), under the guidance of Prof. Anirban Dasgupta.
- **Comprehensive NLP Analysis and Language Modeling** [Aug - Nov 2023]
 - Implemented data scraping from subreddits (e.g., r/india, r/china), sentiment analysis using Hugging Face models, n-gram language modeling with custom smoothing techniques, and pretraining/fine-tuning of BERT for classification and question-answering tasks, demonstrating proficiency in NLP and collaborative teamwork.
- **Online Maintenance Complaint Portal** [Jan - April 2023]
 - Designed a website for IIT Gandhinagar, addressing page reloading, authentication, and security vulnerabilities.
 - Enhanced user experience and fortified the website against potential threats.
- **Machine Learning** [Jan - Oct 2023]
 - Trained a Named Entity Recognition (NER) model using spaCy to extract legal entities from legal documents.
 - Developed a Python library named customized-decision-tree-library for creating decision trees.
 - Created a Streamlit application to visualize the effects of cross-covariance by a 3-sigma ellipse plot.
- **Contact List Search** [BTech Project]
 - Developed a web application for contact management, featuring contact addition and deletion capabilities.
 - Implemented autocomplete functionality for real-time contact name suggestions while typing.

Achievements

- Qualified for the GATE (CS) twice (2021, 2022)
- Solved 350+ questions on LeetCode with a contest ranking of 1447/461,271, as well as additional questions on GeeksForGeeks and CodingNinjas.